BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: MRD-10TH 150523 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section – A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** MRD-10TH 150523 (A)
VERSION: A

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e A, B, C or D.

Q1. The diaphragm is flat, the ribs move out, this occur during.

(A) Expiration
(B) Inspiration
(C) Breathing
(D) Lungs expansion

Q2. The tube between the kidney and urinary bladder is.

(A) Ureter
(B) Urethra
(C) Renal tubule
(D) Nephron

Q3. Secretion of urea in sweat makes skin a.

(A) Neurons
(B) Effectors
(C) Receptors
(D) All of them

Q4. The cones of the eye also contain a pigment called.

(A) Lodospin
(B) Rhodopsin
(C) Blind Spot
(D) Pupil

Q5. Which of these is a part of appendicular skeleton?

(A) Ribs
(B) Sternum
(C) Pectoral girdle
(D) Vertebral column

Q6. In a flower embryosac is formed inside the.

(A) Anther
(B) Ovule
(C) Style
(D) Filament

Q7. The various form of a gene responsible for the same trait are called.

(A) Centromere
(B) Trait
(C) Allele
(D) Chromosome

Q8. Example of tertiary consumer is.

(A) Hawk
(B) Lizard
(C) Frog
(D) Molluscs

Q9. Type of dominance pattern followed by I^A and I^B alleles of human blood group system.

(A) Complete dominance
(B) Incomplete dominance
(C) Co-dominance
(D) None of the above

Q10. All of the following are classes of antibiotics except?

(A) Sulfonamides
(B) Tetracycline
(C) Penicillin
(D) Morphine

Q11. The union of male and female gametes is known as.

(A) Fertilization
(B) Oogenesis
(C) Spermatogenesis
(D) Gametogenesis

Q12. Which connective tissue attaches two bones at a joint?

(A) Tendon
(B) Ligament
(C) Cartilage
(D) Marrow

Section – B

Time Allowed: 2:40 Hours | **Maximum Marks:** 32

Q2. Write short answers to any eight (8) of the following parts. All parts carry equal marks:-

(i) Differentiate between the terms "Respiration" and "Breathing"?

(ii) Draw a labelled diagram of the structure of the "Human Kidney".

(iii) What is the role of insulin and glucagon?

(iv) How do reflex action work in human beings?

(v) What are ligaments and tendons? What function do they perform?

(vi) How can crossing over bring variations?

(vii) Define food chain? Explain with examples?

(viii) Write the characteristics of insect pollinated flowers?

(ix) Outline the role of bacteria in the Nitrogen Cycle?

(x) Write any four advantages of biotechnology in the field of agriculture?

(xi) Define narcotics. Give examples.

Section – C

Maximum Marks: 21

> Note: Answer any three (3) questions. All questions carry equal marks:-

Q3.

(a) Describe "Seed embryo" structure?

(b) Define haemo dialysis. How is it performed?

Q4.

(a) How are different types of arthritis caused? Write their treatment?

(b) Describe three types of neuron on the basis of function?

Q5.

(a) Write the sources of medicinal drugs.

(b) Explain acid rain.

Q6. Define Mendel's law of segregation? Explain the law with the help of an example and chart.